Now using Eq. (5.29), we get

$$\bar{E}_0 = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{N}\int_0^{E_{Fo}} E^{3/2}dE$$

$$= \frac{2}{5}\frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{N}E_{Fo}^{5/2} \tag{5.48}$$

Substituting the value of N from Eq. (5.37), we obtain

$$\bar{E}_0 = \frac{3}{5}E_{Fo} \tag{5.49}$$

This relates the average kinetic energy of an electron at absolute zero to the Fermi energy at the same temperature.

5.3 APPLICATIONS OF THE FREE ELECTRON GAS MODEL

The electronic properties of metals can be broadly divided into the following two groups :

(i) Static properties

(ii) Transport properties

The static properties are those which can be treated effectively by considering just the energy levels or the distribution of energy levels to which the electrons belong. More specifically, we need to consider the overall change in potential of the electrons without investigating deeply the detailed process producing the transitions. These properties include various electron emission properties, the magnetic properties and the properties like heat capacity and contact potential. These properties arise as a result of the excitations of the electrons by light, thermal energy, strong electric fields, etc.

The transport properties are those which can be treated by considering the detailed response of the electrons to an external field, i.e., by considering the acceleration properties of the electrons. These properties include electrical and thermal conductivities, Hall effect, thermoelectricity, etc.

We consider below the electronic specific heat of metals.

5.3.1 Electronic Specific Heat

The average kinetic energy of a free electron as given by the classical statistical mechanics is

$$\bar{E}_0 = \frac{3}{2}k_BT$$

If the metal contains N free electrons, then the total kinetic energy becomes

$$E = N\bar{E}_0 = \frac{3}{2} Nk_BT$$

The electronic specific heat is, therefore, given by

$$C_v = \frac{3}{2} Nk_B \tag{5.50}$$

The measurement of optical reflection coefficient of metals indicates that the number of electrons per atom in metals is of the order of unity. Therefore, considering one gram atom of the metal, the value of electronic specific heat is obtained as (3/2)R or 3 cal $g^{-1}K^{-1}$ approximately. On the other hand, the specific heat associated with the lattice vibrations at high temperatures is about 3R or 6 cal $g^{-1}K^{-1}$. Thus one might conclude that the specific heat of metals should be about 50% greater than the specific heat of insulators. This is contradictory to the experimental observations which indicate that the electronic contribution to the specific heat is very small. At room termperature, the electronic contribution is not more than 0.01 of the value given by Eq. (5.50). Also, this contribution decreases linearly to zero as T approaches zero. It, therefore, appears that all the electrons might not contribute to the specific heat; instead, only a small fraction of the total number of electrons might contribute. This view is supported by the quantum theory.

As described earlier, the quantum mechanics suggests that only those electrons contribute to the specific heat which lie within an energy range k_BT below the Fermi level. This happens because when the electron gas is heated up to a temperature T, these electrons acquire energy of the order of k_BT and jump to the empty higher energy states. The deep lying electrons cannot do so because the unfilled energy states are not available to these electrons for excitation. These electrons, as such, do not contribute to the specific heat. The number of electrons which contribute to the specific heat is of the order $N\left(\frac{k_BT}{E_F}\right)$ or $N\left(\frac{T}{T_F}\right)$ where T_F is called the *Fermi termperature* and is defined by the equation

$$E_F = k_BT_F$$

Using the effective value of N, Eq. (5.50) becomes

$$C_v \cong \frac{3}{2} Nk_B\left(\frac{T}{T_F}\right) \tag{5.51}$$

This indicates that C_v is proportional to T and approaches zero as $T \to 0$. Using typical values of T and T_F as 300 K and 3000 K respectively, the above equation gives the value of C_v as

$$C_v \cong (0.01)\left(\frac{3}{2}\right)Nk_B$$

The specific heat calculated from this equation agrees with the experimental observations. Thus the quantum mechanics modifies the thermal behaviour of free electrons in a simple and satisfying manner.

SOLVED EXAMPLES

Example 5.1. The atomic radius of sodium is 1.86 Å. Calculate the Fermi energy of sodium at absolute zero.

Solution. The Fermi energy at 0 K is given by the relation

$$E_{Fo} = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3}$$

where N is the number of free (valence) electrons present in volume V of the metal. We consider a unit cell of sodium which is *bcc*.

Radius of the sodium atom, $r = 1.86 \times 10^{-10}$m

Volume of the unit cell, $V = a^3 = \left(\frac{4r}{\sqrt{3}}\right)^3$

$$= \left(\frac{4 \times 1.86 \times 10^{-10}}{\sqrt{3}}\right)^3$$

$$= 7.93 \times 10^{-29}\ \text{m}^3$$

Since an atom of sodium has only one valence electron, the number of free electrons in a unit cell of sodium is 2.

$$\therefore \quad \frac{N}{V} = \frac{2}{7.93 \times 10^{-29}} = 2.52 \times 10^{28}\ \text{electrons/m}^3.$$

$$\therefore \quad E_{Fo} = \frac{\left(1.05 \times 10^{-34}\right)^2}{2 \times 9.1 \times 10^{-31}}\left(3\pi^2 \times 2.52 \times 10^{28}\right)^{2/3}$$

$$= 4.98 \times 10^{-19} \text{ J}$$

$$= 3.11 \text{ eV}.$$

Example 5.2. Derive pressure versus volume relationship for a free electron gas at 0 K. Hence obtain an expression for the bulk modulus in terms of the average kinetic energy.

Solution. From thermodynamics, we have

$$P = -\frac{\partial E}{\partial V}$$

where E is the internal energy of a system of particles occupying a volume V at pressure P. For a free electron gas containing N electrons with average kinetic energy $\bar{E}_0$ at 0 K, the energy E may be replaced by $N\bar{E}_0$. Therefore, we have

$$P = -N \frac{\partial \bar{E}_0}{\partial V}$$

Using Eq. (5.49), we obtain

$$P = -\frac{3}{5} N \frac{\partial E_{F0}}{\partial V}$$

Now, from Eq. (5.46), we have

$$E_{F0} = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V} \right)^{2/3}$$

which gives

$$\frac{\partial E_{F0}}{\partial V} = -\frac{2}{3} \frac{\hbar^2}{2m} \left(3\pi^2 N\right)^{2/3} \left(\frac{1}{V}\right)^{5/3}$$

$$\therefore \quad P = \frac{2}{5} N \frac{\hbar^2}{2m} \left(3\pi^2 N\right)^{2/3} \left(\frac{1}{V}\right)^{5/3} \tag{5.52}$$

or

$$P = \frac{2}{5} \frac{NE_{F0}}{V} \tag{5.53}$$

This is the pressure versus volume relationship for a free electron gas at 0 K. The bulk modulus is given by the expression

$$B = -V \frac{\partial P}{\partial V}$$

Differentiating (5.52) with respect to V, we get

$$\frac{\partial P}{\partial V} = -\frac{5}{3}\left[\frac{2}{5}\frac{N\hbar^2}{2m}\left(3\pi^2 N\right)^{2/3}\right]\left(\frac{1}{V}\right)^{7/3}$$

$$= -\frac{5}{3}\frac{P}{V}$$

$$\therefore \qquad B = \frac{5}{3}P = \frac{2}{3}\frac{NE_{Fo}}{V} = \frac{10}{9}\frac{N\bar{E}_o}{V} = \frac{10}{9}\frac{E}{V} \qquad (5.54)$$

where we have used Eqs. (5.53) and (5.49).

SUMMARY

1. According to the free electron gas model, the valence electrons of the atoms present in a metal move freely among the positive ion cores just like the molecules of an ideal gas. These electrons are also called the conduction electrons and are responsible for conduction of electricity.

2. The energy spectrum of a free electron gas present in one or three-dimensional box is discrete particularly when the box has atomic dimensions. For laboratory dimensions, the levels almost form a continuum. The allowed eigen values of the states with wave vector **k** are given as

$$E_k = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right)$$

where m is the mass of the electron.

3. The Fermi level is the level which divides the filled and unfilled states. It is the topmost filled level at 0 K. For T > 0 K, the probability of occupation of this level is 1/2. The position of the Fermi level changes with temperature and is expressed by the relation

$$E_F \cong E_{Fo}\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_{Fo}}\right)^2\right]$$

where E_{Fo} is the Fermi energy at 0 K.

4. The Fermi energy varies with electron concentration n as

$$E_F = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}$$

5. The number of electronic states present in a unit energy range is called the density of states. In a three-dimensional box of volume V, the density of states at energy E is given by

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

At the Fermi level, $D(E)$ may be expressed as

$$D(E_F) = \frac{3}{2}\frac{N}{E_F}$$

where N is the total number of electrons.

6. The average kinetic energy of an electron at 0 K is related to the Fermi energy at the same temperature as

$$\bar{E}_0 = \frac{3}{5} E_{Fo}$$

7. The free electron gas model explains the electronic specific heat of metals by restricting the number of electrons contributing to specific heat to $N\left(\frac{k_B T}{E_F}\right)$. It yielded the expression for specific heat as

$$C_v \cong \frac{3}{2} N k_B \left(\frac{k_B T}{E_F}\right)$$

VERY SHORT QUESTIONS

1. What do you understand by free electron gas ?
2. What is Wiedemann-Franz law ?
3. Define the Fermi energy for metals. How does it depend on mass of the sample?
4. What is the significance of the Fermi distribution function?
5. Define density of states.
6. What do you understand by a Fermi gas ?
7. What is the Fermi sphere ?
8. Can a metal be associated with two Fermi temperatures ?

SHORT QUESTIONS

1. How does a free electron gas differ from an ordinary gas ?
2. How does the classical free electron theory lead to the Ohm's law?
3. Explain complete opacity and high lustre on the basis of the Drude-Lorentz free electron theory.
4. What are the failures of the Drude-Lorentz free electron theory ?
5. Obtain an expression for the energy eigen values of an electron using the Sommerfeld's free electron theory in one dimension.
6. Derive the three-dimensional wave function for the free electron gas and obtain its eigen values.
7. Write the expression for the Fermi-Dirac distribution function and discuss its behaviour with change in temperature.

8. How does the Fermi energy of a metal depend on temperature, concentration of electrons and the total number of electrons present in the metal ?
9. How does the quantum free electron theory explain the observed small values of electronic specific heat of metals ?

LONG QUESTIONS

1. Describe the free electron gas model of metals. How does it help to explain the lattice heat capacity of metals.
2. Obtain expressions for the Fermi energy, the total energy and the density of states for a free electron gas in one dimension. Show the variation of density of states with energy.
3. What is the Fermi energy? Calculate its value for the free electron gas at 0K and mention its significance.
4. Derive expressions for the Fermi energy and density of states for a free elctron gas in one dimension.
5. Show that for a one-dimensional metallic crystal, the average kinetic energy of electron in the ground state is one-third of the Fermi energy.
6. Show that the average kinetic energy per electron for a three-dimensional free electron gas at 0K is

$$\overline{E}_0 = (3/5)E_{F0}$$

where E_{F0} is the Fermi energy at 0K.

7. Discuss qualitatively why the electronic specific heat is temperature dependent and is much less than that expected from the classical behaviour of free electron gas.

PROBLEMS

1. Calculate the Fermi energy of electrons at 0K for a metal with electron density of $10^{28} m^{-3}$. (1.69eV)
2. Silver (*fcc*) has an atomic radius of 1.44 Å. Assuming silver to be monovalent metal, calculate the value of the Fermi energy, the Fermi temperature and the Fermi velocity.
   (5.50 eV, 63850K, 1.39×10^6 ms^{-1})
3. Aluminium metal crystallises in *fcc* structure. If each atom contributes single electron as free electron and the lattice constant is 4Å, calculate (treating conduction electrons as free electron Fermi gas)
   (*i*) Fermi energy (E_F) and the Fermi wave vector (k_F),
   (*ii*) Total kinetic energy of free electron gas per unit volume at 0K.
   Given $\hbar = 1.054 \times 10^{-27}$ erg-s; Electron rest mass = 9.11×10^{-28}g.
   (2.28 eV, 7.73×10^9 m^{-1}, 2.14×10^{28} eVm^{-3})
4. The density of potassium is 860 kgm^{-3}. Calculate the Fermi energy and compressibility of potassium at 0K. (Atomic mass of K = 39.1 amu) (2.02 eV, 3.51×10^{-10} $N^{-1}m^2$)
5. If Fermi temperature is of the order of 5×10^4K, provide a rough estimate of electronic specific heat at 100K. (» 0.025J)